

EDA Sensor-based Evaluation of a Vegetation Succession Learning System

Shuya Kawaguchi, Hiroshi Mizoguchi
Department of Mechanical Engineering
Tokyo University of Science
 2641 Yamazaki, Noda-shi, Chiba,
 Japan
 7517612@ed.tus.ac.jp
 hm@rs.noda.tus.ac.jp

Fusako Kusunoki
Department of Information Design
Tama Art University
 2-1723, Yirimizu, Hachioji-shi, Tokyo,
 Japan
 kusunoki@tamabi.ac.jp

Ryohei Egusa
Department of Education and Child
Development
Meiji Gakuin University
 line 4: City, Country
 egusa@psy.meijigakuin.ac.jp

Hideo Funaoi
Faculty of Education
Soka University
 1-236, Tangi-machi, Hachioji-shi,
 Tokyo, Japan
 funaoi@umegumi.net

Yoshiaki Takeda, Etsuji Yamaguchi,
 Shigenori Inagaki,
Graduate School of Human
Development and Environment
Kobe University
 3-11, Tsurukabuto, Nada-ku, Koube-shi,
 Hyogo, Japan
 takedayo@kobe-u.ac.jp
 etuji@opal.kobe-u.ac.jp
 inagakis@kobe-u.ac.jp

Masanori Sugimoto
Graduate School of Information
Science and Technology
Hokkaido University
 Kita 15, Nishi 8, Kita-ku, Sapporo,
 Hokkaido, Japan
 sugi@ist.hokudai.ac.jp

Abstract—There are many forest problems that need to be solved to move toward a sustainable society; therefore, educating people, especially children, about forest problems is very important. Although schools conventionally educate children through textbooks, children are not expected to learn about forest problems actively from only textbooks. Therefore, this study develops game-type learning material that enables students learn complex mechanisms of vegetation succession and the actual state of succession, while taking interest in forest problems, by performing forest management themselves. For evaluation of the system, we measure the electrodermal activity (EDA) because using a conventional questionnaire can only qualitatively evaluate whether learning using the system is more interesting than when using text. EDA sensors are used to measure physiological response of participants reading texts and those using the system; the interest when using the system is evaluated quantitatively. The experiment shows that the change of EDA increases when using the system, which suggests that using the system is more interesting and can be better for learning than reading text.

Keywords— *Electrodermal Activity, SCR, Active Learning, Forest management game*

I. INTRODUCTION

Currently, we need to solve many forest problems around the world and move toward a sustainable society on a global scale [1]. To that end, educating children about forest problems is essential; children should be aware about forests, which is a natural environment, that enables a sustainable society. To build responsible sustainable society in the future, it is important to cultivate skills and attitude related to forests, sensibility, sociality, ability to solve problems, and develop human resources. However, there are challenges in teaching children about forest problems. For children to learn about forest problems and enrich their sensibility and social nature

associated with forests, it is necessary to explain complicated mechanisms such as vegetation succession, which is one of the key concepts in understanding forests, and the realities of succession. Moreover, it is necessary for children to be interested in forest problems and learn them actively. However, children who do not have sufficient information cannot actively learn about forest problems, and forest problem education makes it difficult to enrich the sensibility and sociability associated with forests. It is considered to be. In schools, conventionally, forest problems are taught using textbooks; however, children cannot actively learn about forest problems only from text, due to which adequate education is not possible. To solve this problem, we develop game-type learning material that enables students to learn about complicated mechanisms; this system allows students to manage forests themselves so that they remain interested in learning about forest problems [2, 3].

Using this game-type learning material, children can manage a forest in a virtual environment. Forests in this game are those used by humans for obtaining fuels, firewood, and charcoal, which results in repeated logging and regeneration in the forests; as the succession progresses in the natural state, it becomes necessary to manage these forests because the forest ecosystem is disrupted.

In this paper, we outline the game-type learning material for forest problem education developed by the authors. In addition, in order to evaluate it, we describe preliminary experiments and results obtained from college students. Questionnaires, which are one way of evaluating such a system, can only provide qualitative evaluation. Therefore, we used sensors that can measure physiological responses, i.e., electrodermal activity (EDA), to determine whether learners are interested in forest problems and are motivated by the system; their interests are quantitatively evaluated.

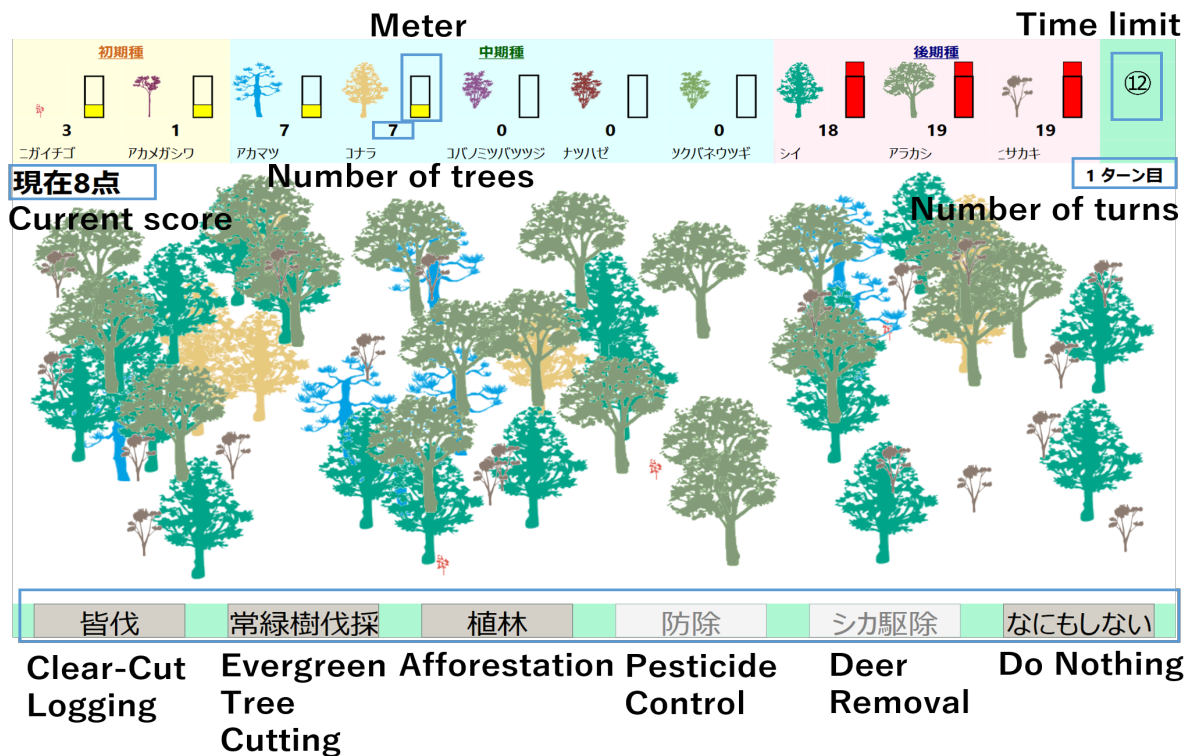


Fig. 1. Initial game screen

II. VEGETATION SUCCESSION LEARNING SUPPORT SYSTEM

In this research, games are developed to support learning complicated mechanisms such as vegetation succession and the actual state of the succession while maintaining interest in forest problems. The developed forest management game allows managing one forest in a virtual environment for approximately 300 years. Because this game can be played on a personal computer, it can be used not only in school classrooms but also in various other places. In this game, ten types of trees are planted in the forest, and they are divided into early-stage species, middle-stage species, and late-stage species. Learners initially improve forests starting in an adverse environment (environments with poor biodiversity = large-stage species, which is a plant with a biodiversity = many vegetation) to an ideal state (an environment rich in biodiversity) I have to do. Fig. 1 shows the initial state of the game. Because each tree has different size and growth rate, a superiority relationship exists. Moreover, there are deer and insects that eat specific trees; therefore, damage due to deer and insects should also be considered, as shown in Fig. 2. The learner manages this forest using the following six management methods: Clear-Cut Logging, Evergreen Tree Cutting, Afforestation, Pesticide Control, Deer Removal, and Do Nothing. Plants that are vegetating change due to their superiority/inferiority relationships with other plants, damage by deer and insects, and influence by people. If the learner selects one management method within the time limit of 15 s per turn, the game proceeds to the next turn and a change corresponding to the action occurs in the forest; this is repeated 20 times. Thus, ways to improve forests by managing the changes in forests due to various factors can be learnt through trial and error. At the same time, it is possible to learn about complicated mechanisms such as vegetation succession and the actual state of succession.

The following functions are provided to assist learner management: score display of the influence of the superiority/inferiority relationship among plants, influence of deer and damage by insects, influence of management by man, and managed forests. With the ideal state considered as 100 points, the current forest state is always displayed as a score. Evergreen trees are constantly increasing. As a result, short trees cannot be planted and the forests will have lower biodiversity. That is, if people do not manage the forests well, the score finally becomes 0, as shown in Fig. 3. Second, to determine the ideal number for each plant, the state of the current number of the plant relative to the ideal number is indicated by a meter. Because the ideal number is different for each plant, there are meters for each plant. When the meter is full, it implies that the ideal number is attained, and when it exceeds the limit, it indicates that there are more trees of that type than necessary. With these functions, learners can learn how to manage forests. Fig. 4 shows the forest managed by a learner.

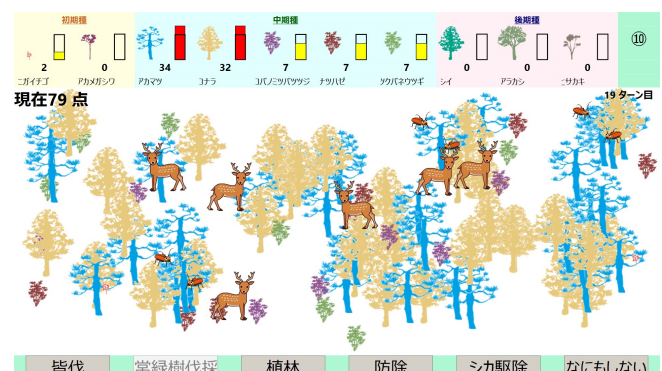


Fig. 2. Damage by deer and insects

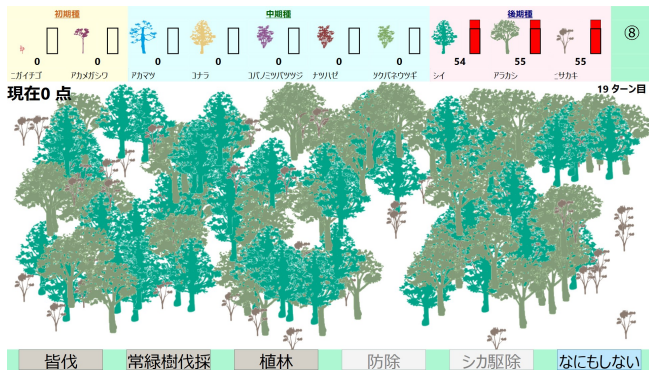


Fig. 3. Forest not managed by a player

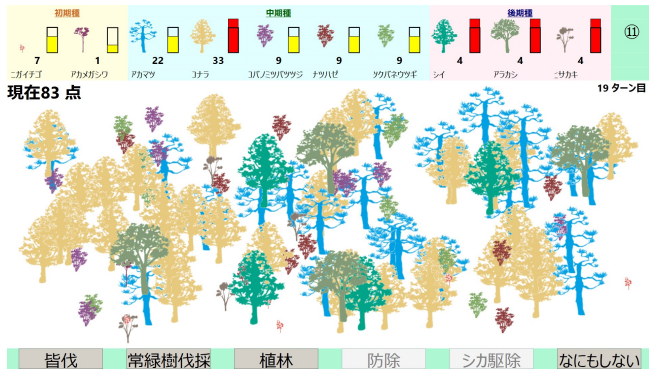


Fig. 4. Forest managed by a player

III. EVALUATION

A. Objective

In this research, we developed a system for learners to focus on forest management in the virtual environment and learn about complicated mechanisms such as vegetation succession and the actual state of succession while maintaining interest in forest problems. We assumed that this environment and such activities stimulate learners' interest. However, it cannot be confirmed whether learning using forest management games is more interesting than learning using text alone. Therefore, we examined whether children are more interested in forest management games. We quantitatively assessed a child's interest using the physiological response, i.e., electrodermal activity (EDA). Using this method, we aimed to evaluate the usefulness of our system.

B. Evaluation by EDA

We confirmed the interest of children using EDA. EDA shows changes in the electrical properties of the skin surface due to sweating caused by excitation or tension. The method of measuring the apparent resistance by passing electricity through the skin is called the energization method. The sustained activity measured thereby is called the skin conductance level (SCL); the temporary activity is called skin conductance response (SCR).

The limbic system of the brain consists of cingulate gyrus, hippocampus, and amygdala. Recent studies have shown that the limbic system is activated when people exhibit interest. Furthermore, it is known that this system is closely related to emotions [4]. When people are visually interested in objects, they experience a pleasant sensation and an increase in arousal, change in skin's electrical activity,

and produce SCR [5]. When interest is generated, SCL and SCR reaction values can be measured by the variations over relatively long and short periods, respectively. When concern arises, a reaction appears. In contrast, if a person is not interested, no response will occur. Therefore, we used EDA to determine if a person is interested in the proposed system. The EDA sensor is shown in Fig. 5.

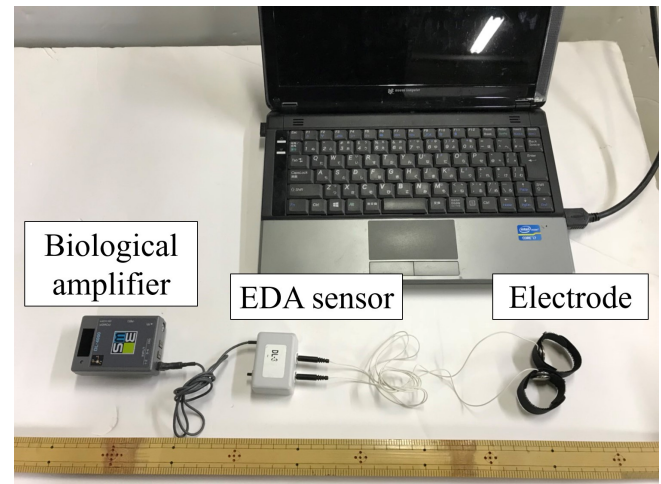


Fig. 5. Measuring device

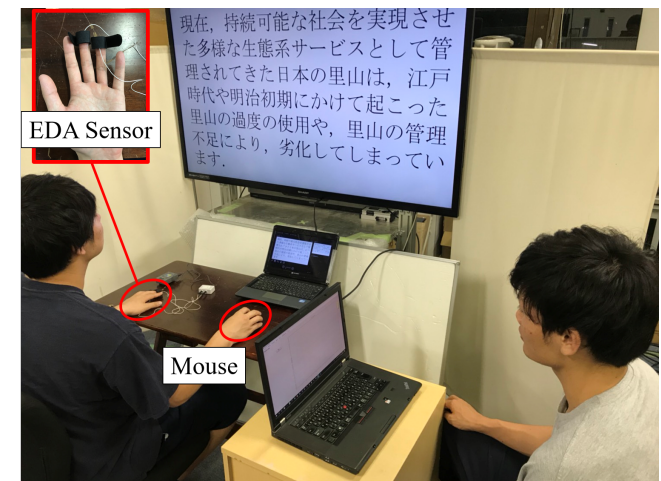


Fig. 6. Experimental environment (reading text)

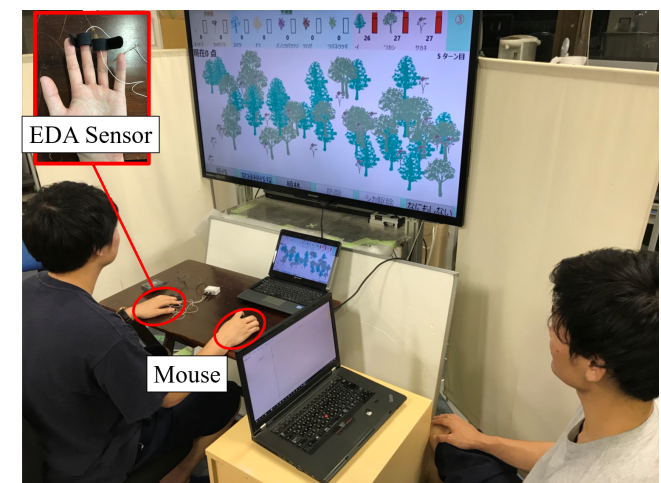


Fig. 7. Experimental environment (using the system)

C. Method

Ten students from Tokyo University of Science in Chiba prefecture, Japan, participated in the experiment. In this experiment, we compared the traditional learning method using text and learning using the proposed forest management game. Then, we evaluated the interest of the learners. For a fair comparison, in the proposed method, a mouse was used to manipulate forests and in the conventional method, the text was assumed to be read through mouse operation as well. The contents of the text summarized everything that can be learned from the forest management game. In the experiment, the ten participants were divided into two groups (A, B) of five people. Group A first experienced the conventional method and then the proposed method, and vice versa for Group B. During the experiment, EDA sensors were attached to the fingers of the participant's left hand and EDA measurements were obtained. Then, the variation per unit time [$\mu\Omega / s$] was calculated for each period, and this difference was evaluated by the t-test. Figs. 6 and 7 show the experimental environment.

D. Result

Fig. 8 shows the experimental results for one subject. The blue line represents the change in SCR when reading the text. The red line represents the change in the SCR when experiencing the system. Tab. 1 shows the value of the fluctuation for all subjects. The results of calculations and comparisons of the average of the skin electric activity for the ten participants are shown in Fig. 9. The results were compared using the t-test. It was confirmed that the variation of electrodermal activity was significantly larger when using the system ($P < 0.05$). From this, it was suggested that interest is improved when learning using the system.

IV. CONCLUSIONS

In this study, we outlined the proposed vegetation succession learning support system and the EDA measurement experiment conducted for its evaluation. We evaluated learning using the system by measuring EDA while reading text and using the system. As a result, it was confirmed that the amount of change in skin electrical activity was significantly larger when using the system. This suggests that learning using the system improves interest in vegetation succession.

In the future, we will measure the gaze of the learner who is experiencing the game with a sensor. Thereby evaluating this game.

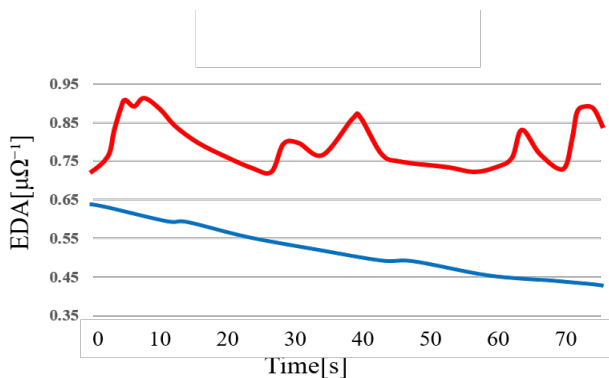


Fig. 8. Experimental result for one subject

TABLE I. RESULTS FOR ALL SUBJECTS

	Subject	Variation of EDA [$\mu\Omega/s$]	
		Text	System
A	1	0.003407667	0.005411634
	2	0.004297256	0.006183495
	3	0.009497163	0.011612181
	4	0.003810646	0.005324506
	5	0.002664637	0.003207735
B	6	0.003969048	0.004419132
	7	0.005741268	0.007023947
	8	0.002761476	0.002633021
	9	0.004577218	0.004404195
	10	0.004401933	0.002790470
Average		0.004512831	0.005301031

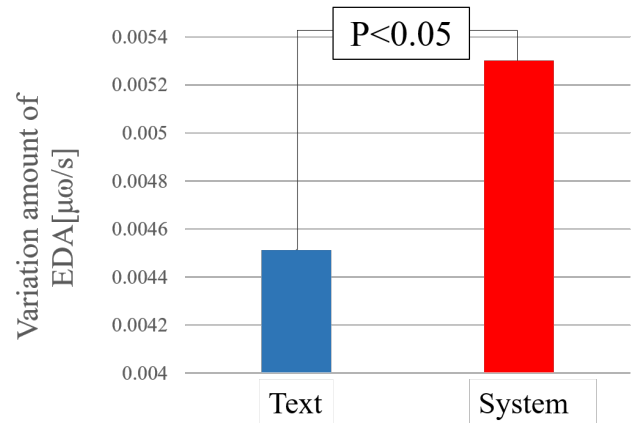


Fig. 9. SCR t-test result

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